Session 15 Quiz - AI That Sees - Computer Vision

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57373]
Computer Vision is defined as AI that works specifically with pictures and _____.

A. Paragraphs
B. Audio files
C. Videos
D. Spreadsheets

Correct Answer: C. Videos

Explanation: Computer Vision systems are designed to interpret visual information from both still images (pictures) and moving footage (videos). Paragraphs are text, audio files are sound, and spreadsheets are data tables, none of which are visual media.

Q2 [Type: MCQ | Difficulty: Easy | QID:57374]
Each pixel in a digital image is represented by three numbers: Red, _____, and Blue.

A. White
B. Yellow
C. Orange
D. Green

Correct Answer: D. Green

Explanation: Digital images use the RGB (Red, Green, Blue) color model. These three colors of light combine at different intensities to create every other color on your screen.

Q3 [Type: MCQ | Difficulty: Easy | QID:57375]
The three-step recipe for most Computer Vision systems is See, _____, and Act.

A. Delete
B. Draw
C. Process
D. Listen

Correct Answer: C. Process

Explanation: The "Process" step is the middle phase where the AI analyzes patterns like edges, shapes, and textures to understand the image. Delete, Draw, and Listen are not part of the standard Computer Vision recipe.

Q4 [Type: MCQ | Difficulty: Easy | QID:57376]
A standard photo taken on a modern smartphone consists of roughly 12 _____ pixels.

A. Million
B. Thousand
C. Trillion
D. Billion

Correct Answer: A. Million

Explanation: Most phone cameras capture about 12 megapixels, which translates to 12 million individual colored dots. "Mega" means million. Thousand, trillion, and billion are not correct for standard smartphone photos.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:57377]
When a computer identifies and reads text found inside an image, it is performing a job known as _____.

A. OCR
B. GPS
C. NLP
D. RGB

Correct Answer: A. OCR

Explanation: Optical Character Recognition (OCR) is the specific Computer Vision job dedicated to turning images of text (like scanned documents or signs) into actual machine-readable text data.

Q6 [Type: True/False | Difficulty: Easy | QID:57378]
True or False: AI sees a human face as a single, complete object rather than a large collection of individual squares.

A. True
B. False

Correct Answer: B. False

Explanation: To an AI, an image is just a grid of millions of pixels (individual squares). It must find patterns within those squares to identify a face. It does not see the face as a single whole object like a human does.

Q7 [Type: True/False | Difficulty: Easy | QID:57379]
True or False: In the RGB color system, a value of (0,0,0) represents the color pure white.

A. False
B. True

Correct Answer: A. False

Explanation: A value of (0,0,0) means there is no red, green, or blue light, which results in pure black. Pure white would be (255,255,255), the maximum intensity of all three colors.

Q8 [Type: True/False | Difficulty: Easy-Medium | QID:57380]
True or False: 'Occlusion' is a term used when an object is partially hidden or covered, such as someone wearing a mask.

A. True
B. False

Correct Answer: A. True

Explanation: Occlusion makes Computer Vision harder because the AI loses important visual clues it uses for identification. Masks, hands over the face, or objects blocking the view are all examples of occlusion.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:57381]
True or False: Face recognition systems always work with the same level of accuracy for every person, regardless of who was in the training data.

A. True
B. False

Correct Answer: B. False

Explanation: Bias in training data can lead to systems that are less accurate for certain groups. For example, if a face recognition system is trained mostly on light-skinned faces, it will be less accurate for darker-skinned faces.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:57382]
True or False: Identifying exactly where objects are by drawing boxes around them is a job called Object Detection.

A. True
B. False

Correct Answer: A. True

Explanation: Object Detection focuses on locating specific items within a scene. It answers "Where are the things?" by drawing bounding boxes around objects like cars, people, or stop signs, rather than just classifying the whole image.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57383]
In the 3-step recipe for Computer Vision, which step specifically involves the camera capturing the millions of pixels?

A. See
B. Act
C. Process
D. Learn

Correct Answer: A. See

Explanation: The "See" step is the input phase where the computer takes in raw pixel data from the environment via a camera. It is similar to how your eyes capture light before your brain processes the image.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57384]
If an individual pixel has the RGB values (255,255,255), what color does it display on your screen?

A. Transparent
B. White
C. Black
D. Gray

Correct Answer: B. White

Explanation: When red, green, and blue light are all at their maximum intensity (255), they combine to create pure white. (0,0,0) is black, and equal lower values like (128,128,128) give gray.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57385]
Why might a self-driving car struggle to correctly identify a stop sign during a heavy blizzard?

A. The car is moving too slowly
B. The computer gets too cold to think
C. The stop sign changes its RGB values
D. The sign is partially hidden by snow

Correct Answer: D. The sign is partially hidden by snow

Explanation: Physical barriers like snow can hide the distinct shape and color patterns the AI needs to recognize the sign. This is occlusion. The sign's color does not change, and computer operating temperature is not the issue.

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57386]
Which Computer Vision job is being used when Google Photos automatically groups all your vacation pictures into a folder titled 'Beach Day'?

A. OCR
B. Image Classification
C. Anomaly Detection
D. Object Detection

Correct Answer: B. Image Classification

Explanation: Image Classification answers the question "What is this whole image of?" by assigning a category like "beach," "cat," or "birthday party." OCR reads text, anomaly detection finds outliers, and object detection locates specific items.

Q15 [Type: MCQ | Difficulty: Medium | QID:57387]
When an AI processes an image to find objects, what is typically the very first level of patterns it looks for?

A. Textual labels
B. Familiar faces
C. The meaning of the photo
D. Edges

Correct Answer: D. Edges

Explanation: AI starts by finding where colors suddenly change (edges), which helps it create an outline of objects. Only after finding edges can it detect shapes, then textures, then whole objects. Edges are the most basic visual pattern.

Q16 [Type: MCQ | Difficulty: Easy | QID:57388]
What does the 'B' stand for in the RGB color model used by computers to see?

A. Black
B. Brown
C. Blue
D. Bright

Correct Answer: C. Blue

Explanation: RGB stands for Red, Green, Blue, the three primary colors of light used in digital displays. Black, brown, and bright are not part of this acronym.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57389]
Which of these is a common reason why a face unlock feature might fail to recognize its owner?

A. The phone is being held at a very sharp angle
B. The battery is at 100%
C. The owner is wearing a different shirt
D. The owner is standing in a quiet room

Correct Answer: A. The phone is being held at a very sharp angle

Explanation: Angles can dramatically change how a face looks to an AI, making it hard to match with its learned patterns. Wearing a different shirt usually does not affect face recognition; battery and room noise are irrelevant.

Q18 [Type: MCQ | Difficulty: Medium | QID:57390]
According to the source material, why were some face recognition tools significantly less accurate for dark-skinned women?

A. Cameras cannot see dark colors
B. The training data mostly contained light-skinned faces
C. The AI was not powerful enough
D. Dark-skinned women use more Snapchat filters

Correct Answer: B. The training data mostly contained light-skinned faces

Explanation: AI learns from examples. If its training data lacks diversity (mostly light-skinned faces), it will not learn to recognize darker skin tones equally well. This is a well-documented bias in early face recognition systems.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57391]
A self-driving car identifying a red light and then physically stopping is an example of which step in the CV recipe?

A. See
B. Recall
C. Act
D. Process

Correct Answer: C. Act

Explanation: The "Act" step is the final stage where the computer actually does something based on what it saw and processed. Here, the car stops. "See" is capturing the image, "Process" is identifying the red light, and "Act" is braking.

Q20 [Type: MCQ | Difficulty: Medium | QID:57392]
Which specific Computer Vision job would be most helpful for a doctor trying to find a tiny bruise on a piece of fruit or a tumor on an X-ray?

A. Anomaly Detection
B. Snapchat Filters
C. Face Recognition
D. OCR

Correct Answer: A. Anomaly Detection

Explanation: Anomaly Detection is designed to spot patterns that look "wrong" or different from what is normal or healthy. A bruise on fruit or a tumor on an X-ray are anomalies (unexpected variations) that this job can highlight.